

CHAPTER ONE

KINEMATICS OF A PARTICLE: RECTILINEAR KINEMATICS

Learning Objectives

After completing this chapter, students should be able to do the following:

- Distinguish the various classifications of rigid-body mechanics
- Analyse and solve equations of kinematics of particles undergoing rectilinear motion

1.1 Overview of Mechanics

Mechanics is a branch of the physical sciences that is concerned with the state of rest or motion of bodies that are subjected to the action of forces. In general, this subject can be subdivided into three branches: **rigid-body mechanics**, **deformable-body mechanics or solid mechanics**, and **fluid mechanics**. The study of rigid-body mechanics is a basic requirement for the study of the mechanics of deformable bodies and the mechanics of fluids. Furthermore, rigid-body mechanics is essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering.

1.2 Overview of Rigid-body Mechanics

Rigid-body mechanics is divided into two areas: (1) Statics and (2) Dynamics.

1.2.1 **Statics**: It is that branch of Engineering Mechanics, which deals with the forces and their effects, while acting upon the bodies at rest. We can consider statics as a special case of dynamics, in which the acceleration is zero.

1.2.2 **Dynamics**: It is that branch of Engineering Mechanics, which deals with the forces and their effects, while acting upon the bodies in motion. The subject of Dynamics may be further subdivided into the following two branches: (1) Kinetics and (2) Kinematics.

(1) **Kinetics**: It is the branch of Dynamics, which deals with the bodies in motion due to the application of forces.

(2) **Kinematics**: It is that branch of Dynamics, which deals with the bodies in motion, without any reference to the forces which are responsible for the motion.

1.3 Rectilinear Kinematics: Continuous Motion

Rectilinear Kinematics can be described as a **straight-lined motion**. The kinematics of a particle is characterized by specifying, at any given instant, the particle's **position**, **velocity**, and **acceleration**.

1.3.1 Position

The straight-line path of a particle will be defined using a single coordinate axis s , Fig. 1.1a. The origin O on the path is a fixed point, and from this point the position coordinate s is used to specify the location of the particle at any given instant. The magnitude of s is the distance from O to the particle, usually measured in meters (m) or feet (ft), and the sense of direction is defined by the algebraic sign on s .

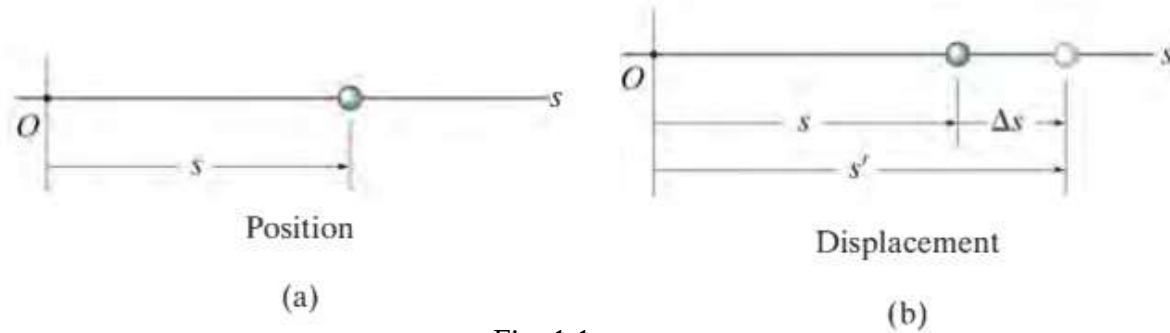


Fig. 1.1

1.3.2 Displacement

The displacement of the particle is defined as the change in its position. For example, if the particle moves from one point to another, Fig. 1.1b, the displacement is

$$\Delta s = s' - s$$

In this case Δs is positive since the particle's final position is to the right of its initial position, i.e., $s' > s$. Likewise, if the final position were to the left of its initial position, Δs would be negative. The displacement of a particle is also a vector quantity, and it should be distinguished from the distance the particle travels. Specifically, the distance travelled is a positive scalar that represents the total length of path over which the particle travels.

1.3.3 Velocity

If the particle moves through a displacement Δs during the time interval Δt , the average velocity of the particle during this time interval is

$$v_{avg} = \frac{\Delta s}{\Delta t}$$

If we take smaller and smaller values of Δt , the magnitude of Δs becomes smaller and smaller. Consequently, the *instantaneous velocity* is a vector defined as $v = \lim_{\Delta t \rightarrow 0} (\Delta s / \Delta t)$, or

$$\left(\begin{smallmatrix} + \\ \rightarrow \end{smallmatrix} \right) \quad v = \frac{ds}{dt} \quad (1.1)$$

Since Δt or dt is always positive, the sign used to define the sense of the velocity is the same as that of Δs or ds . The average speed is always a positive scalar and is defined as the total distance traveled by a particle, s_T , divided by the elapsed time Δt ; i.e

$$(v_{sp})_{avg} = \frac{s_T}{\Delta t}$$

1.3.4 Acceleration

Provided the velocity of the particle is known at two points, the average acceleration of the particle during the time interval Δt is defined as

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

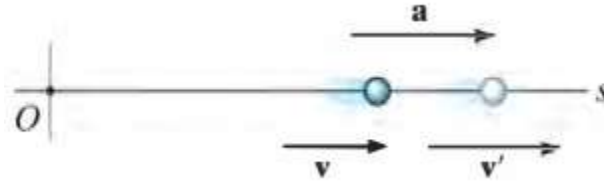


Fig. 1.2

Here Δv represents the difference in the velocity during the time interval Δt , i.e., Fig. 1.2.

$$\Delta v = v' - v$$

The *instantaneous acceleration* at time t is a vector that is found by taking smaller and smaller values of Δt and corresponding smaller and smaller values of Δv , so that $a = \lim_{\Delta t \rightarrow 0} (\Delta v / \Delta t)$, or

$$\left(\begin{smallmatrix} + \\ \rightarrow \end{smallmatrix} \right) \quad a = \frac{dv}{dt} \quad (1.2)$$

Substituting Eq. 1.1 into this result, we can also write

$$\left(\begin{smallmatrix} + \\ \rightarrow \end{smallmatrix} \right) \quad a = \frac{d^2s}{dt^2}$$

Both the average and instantaneous acceleration can be either positive or negative.

Finally, an important differential relation involving the displacement, velocity, and acceleration along the path may be obtained by eliminating the time differential Δt between Eqs. 1.1 and 1.2, which gives

$$\left(\overset{+}{\rightarrow}\right) \quad a ds = v dv \quad (1.3)$$

Although we have now produced three important kinematic equations, realize that the above equation is not independent of Eqs. 1.1 and 1.2.

1.3.5 Constant Acceleration

Constant acceleration, $a = a_c$. When the acceleration is constant, each of the three kinematic equations $a_c = dv/dt$, $v = ds/dt$, and $a_c ds = v dv$ can be integrated to obtain formulas that relate a_c, v, s , and t .

1.3.6 Velocity as a Function of Time

Integrate $a_c = dv/dt$, assuming that initially $v = v_0$ when $t = 0$.

$$\left(\overset{+}{\rightarrow}\right) \quad \int_{v_0}^v dv = a_c \int_0^t dt$$

$$v = v_0 + a_c t \quad (1.4)$$

1.3.7 Position as a Function of Time

Integrate $v = ds/dt = v_0 + a_c t$, assuming that initially $s = s_0$, when $t = 0$.

$$\int_{s_0}^s ds = \int_0^t (v_0 + a_c t) dt$$

$$\left(\overset{+}{\rightarrow}\right) \quad s = s_0 + v_0 t + \frac{1}{2} a_c t^2 \quad (1.5)$$

1.3.8 Velocity as a Function of Position

Either solve for t in Eq. 1.4 and substitute into Eq. 1.5, or integrate $v dv = a_c ds$, assuming that initially $v = v_0$ when $s = s_0$.

$$\int_{v_0}^v v dv = \int_{s_0}^s a_c ds$$

$$\left(\overset{+}{\rightarrow}\right) \quad v^2 = v_0^2 + 2a_c(s - s_0) \quad (1.6)$$

The algebraic signs of s_0 , v_0 , and a_c , used in the above three equations, are determined from the positive direction of the s axis as indicated by the arrow written at the left of each equation. Remember that these equations are useful only when the acceleration is constant and when $t = 0$, $s = s_0$, $v = v_0$.

Example 1.1

The car in Fig. E-1.1 moves in a straight line such that for a short time its velocity is defined by $v = (3t^2 + 2t) \text{ ft/s}$, where t is in seconds. Determine its position and acceleration when $t = 3 \text{ s}$. When $t = 0$, $s = 0$.



Fig. E-1.1

SOLUTION

Coordinate System. The position coordinate extends from the fixed origin O to the car, positive to the right.

Position. Since $v = f(t)$, the car's position can be determined from $v = ds/dt$, since this equation relates v , s , and t . Noting that $s = 0$ when $t = 0$, we have*

$$(\rightarrow) \quad v = \frac{ds}{dt} = (3t^2 + 2t)$$

$$\int_0^s ds = \int_0^t (3t^2 + 2t) dt$$

$$s \Big|_0^s = t^3 + t^2 \Big|_0^t$$

$$s = t^3 + t^2$$

When $t = 3$ s,

$$s = (3)^3 + (3)^2 = 36 \text{ ft} \quad \text{Ans.}$$

Acceleration. Since $v = f(t)$, the acceleration is determined from $a = dv/dt$, since this equation relates a , v , and t .

$$\begin{aligned} (\rightarrow) \quad a &= \frac{dv}{dt} = \frac{d}{dt}(3t^2 + 2t) \\ &= 6t + 2 \end{aligned}$$

When $t = 3$ s,

$$a = 6(3) + 2 = 20 \text{ ft/s}^2 \rightarrow \quad \text{Ans.}$$

NOTE: The formulas for constant acceleration *cannot* be used to solve this problem, because the acceleration is a function of time.

*The *same result* can be obtained by evaluating a constant of integration C rather than using definite limits on the integral. For example, integrating $ds = (3t^2 + 2t)dt$ yields $s = t^3 + t^2 + C$. Using the condition that at $t = 0$, $s = 0$, then $C = 0$.

Example 1.2

A small projectile is fired vertically downward into a fluid medium with an initial velocity of 60 m/s. Due to the drag resistance of the fluid the projectile experiences a deceleration of $a = (-0.4v^3) \text{ m/s}^2$, where v is in m/s. Determine the projectile's velocity and position 4 s after it is fired.

SOLUTION

Coordinate System. Since the motion is downward, the position coordinate is positive downward, with origin located at 0, Fig. E-1.2.



Fig. E-1.2

Velocity. Here $a = f(v)$ and so we must determine the velocity as a function of time using $a = dv/dt$, since this equation relates v , a , and t . (Why not use $v = v_0 + a_c t$?) Separating the variables and integrating, with $v_0 = 60 \text{ m/s}$ when $t = 0$, yields

$$\begin{aligned}
 (+\downarrow) \quad a &= \frac{dv}{dt} = -0.4v^3 \\
 \int_{60 \text{ m/s}}^v \frac{dv}{-0.4v^3} &= \int_0^t dt \\
 \frac{1}{-0.4} \left(\frac{1}{-2} \right) \frac{1}{v^2} \bigg|_{60}^v &= t - 0
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{0.8} \left[\frac{1}{v^2} - \frac{1}{(60)^2} \right] &= t \\
 v &= \left\{ \left[\frac{1}{(60)^2} + 0.8t \right]^{-1/2} \right\} \text{ m/s}
 \end{aligned}$$

Here the positive root is taken, since the projectile will continue to move downward. When $t = 4 \text{ s}$,

$$v = 0.559 \text{ m/s} \downarrow \quad \text{Ans.}$$

Position. Knowing $v = f(t)$, we can obtain the projectile's position from $v = ds/dt$, since this equation relates s , v , and t . Using the initial condition $s = 0$, when $t = 0$, we have

$$\begin{aligned}
 (+\downarrow) \quad v &= \frac{ds}{dt} = \left[\frac{1}{(60)^2} + 0.8t \right]^{-1/2} \\
 \int_0^s ds &= \int_0^t \left[\frac{1}{(60)^2} + 0.8t \right]^{-1/2} dt \\
 s &= \frac{2}{0.8} \left[\frac{1}{(60)^2} + 0.8t \right]^{1/2} \bigg|_0^t \\
 s &= \frac{1}{0.4} \left\{ \left[\frac{1}{(60)^2} + 0.8t \right]^{1/2} - \frac{1}{60} \right\} \text{ m}
 \end{aligned}$$

When $t = 4 \text{ s}$,

$$s = 4.43 \text{ m} \quad \text{Ans.}$$

Example 1.3

During a test a rocket travels upward at 75 m/s, and when it is 40 m from the ground its engine fails. Determine the maximum height s_B reached by the rocket and its speed just before it hits the ground. While in motion the rocket is subjected to a constant downward acceleration of 9.81 m/s^2 due to gravity. Neglect the effect of air resistance.

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Coordinate System. The origin 0 for the position coordinate s is taken at ground level with positive upward, Fig. E-1.3.

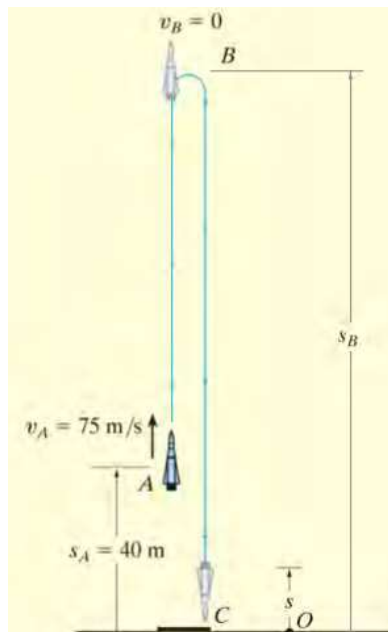


Fig. E-1.3

Maximum Height. Since the rocket is traveling *upward*, $v_A = +75 \text{ m/s}$ when $t = 0$. At the maximum height $s = s_B$ the velocity $v_B = 0$. For the entire motion, the acceleration is $a_c = -9.81 \text{ m/s}^2$ (negative since it acts in the *opposite* sense to positive velocity or positive displacement). Since a_c is *constant* the rocket's position may

be related to its velocity at the two points A and B on the path by using Eq. 1.6, namely

$$\begin{aligned} (+\uparrow) \quad v_B^2 &= v_A^2 + 2a_c(s_B - s_A) \\ 0 &= (75 \text{ m/s})^2 + 2(-9.81 \text{ m/s}^2)(s_B - 40 \text{ m}) \\ s_B &= 327 \text{ m} \end{aligned}$$

Ans.

Velocity: To obtain the velocity of the rocket just before it hits the ground, we can apply Eq. 1-6 between points B and C, Fig. E-1.3.

$$\begin{aligned}
 (+\uparrow) \quad v_C^2 &= v_B^2 + 2a_c(s_C - s_B) \\
 &= 0 + 2(-9.81 \text{ m/s}^2)(0 - 327 \text{ m}) \\
 v_C &= -80.1 \text{ m/s} = 80.1 \text{ m/s} \downarrow \quad \text{Ans.}
 \end{aligned}$$

The negative root was chosen since the rocket is moving downward. Similarly, Eq. 1.6 may also be applied between points A and C, i.e.,

$$\begin{aligned}
 (+\uparrow) \quad v_C^2 &= v_A^2 + 2a_c(s_C - s_A) \\
 &= (75 \text{ m/s})^2 + 2(-9.81 \text{ m/s}^2)(0 - 40 \text{ m}) \\
 v_C &= -80.1 \text{ m/s} = 80.1 \text{ m/s} \downarrow \quad \text{Ans.}
 \end{aligned}$$

NOTE: It should be realized that the rocket is subjected to a *deceleration* from A to B of 9.81 m/s^2 , and then from B to C it is *accelerated* at this rate. Furthermore, even though the rocket momentarily comes to *rest* at B ($v_B = 0$) the acceleration at B is still 9.81 m/s^2 downward!

Example 1.4

A metallic particle is subjected to the influence of a magnetic field as it travels downward through a fluid that extends from plate A to plate B, Fig. E-1.4. If the particle is released from rest at the midpoint C, $s = 100 \text{ mm}$, and the acceleration is $a = (4s) \text{ m/s}^2$, where s is in meters, determine the velocity of the particle when it reaches plate B, $s = 200 \text{ mm}$, and the time it takes to travel from C to B.

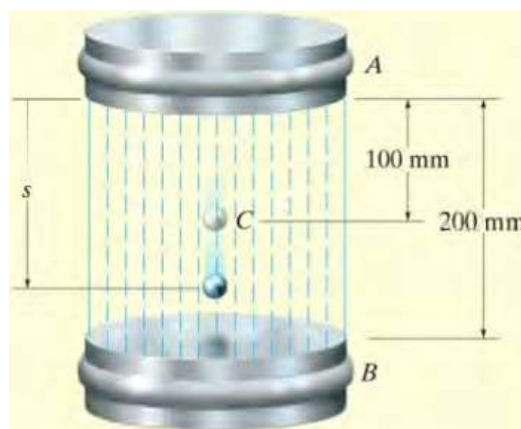


Fig. E-1.4

SOLUTION

Coordinate System. As shown in Fig. E-1.4, s is positive downward, measured from plate A.

Velocity. Since $a = f(s)$, the velocity as a function of position can be obtained by using $v dv = a ds$. Realizing that $v = 0$ at $s = 0.1$ m, we have

$$(+\downarrow) \quad v dv = a ds$$

$$\int_0^v v dv = \int_{0.1 \text{ m}}^s 4s ds$$

$$\left. \frac{1}{2} v^2 \right|_0^v = \left. \frac{4}{2} s^2 \right|_{0.1 \text{ m}}^s$$

$$v = 2(s^2 - 0.01)^{1/2} \text{ m/s} \quad (1)$$

At $s = 200 \text{ mm} = 0.2 \text{ m}$,

$$v_B = 0.346 \text{ m/s} = 346 \text{ mm/s} \downarrow \quad \text{Ans.}$$

The positive root is chosen since the particle is traveling downward, i.e., in the $+s$ direction.

Time. The time for the particle to travel from C to B can be obtained using $v = ds/dt$ and Eq. 1, where $s = 0.1 \text{ m}$ when $t = 0$. From Appendix A,

$$(+\downarrow) \quad ds = v dt$$

$$= 2(s^2 - 0.01)^{1/2} dt$$

$$\int_{0.1}^s \frac{ds}{(s^2 - 0.01)^{1/2}} = \int_0^t 2 dt$$

$$\ln(\sqrt{s^2 - 0.01} + s) \Big|_{0.1}^s = 2t \Big|_0^t$$

$$\ln(\sqrt{s^2 - 0.01} + s) + 2.303 = 2t$$

At $s = 0.2 \text{ m}$,

$$t = \frac{\ln(\sqrt{(0.2)^2 - 0.01} + 0.2) + 2.303}{2} = 0.658 \text{ s} \quad \text{Ans.}$$

Note: The formulas for constant acceleration cannot be used here because the acceleration changes with position, i.e., $a = 4s$.

Example 1.5

A particle moves along a horizontal path with a velocity of $v = (3t^2 - 6t) \text{ m/s}$, where t is the time in seconds. If it is initially located at the origin 0, determine the distance traveled in 3.5s, and the particle's average velocity and average speed during the time interval.

SOLUTION

Coordinate System. Here positive motion is to the right, measured from the origin 0, Fig. E-1.5a.

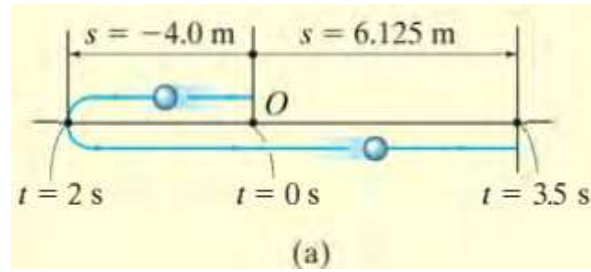


Fig. E-1.5a

Distance Traveled. Since $v = f(t)$, the position as a function of time may be found by integrating $v = ds/dt$ with $t = 0, s = 0$.

($\Rightarrow$)

$$ds = v dt$$

$$= (3t^2 - 6t)dt$$

$$\int_0^s ds = \int_0^t (3t^2 - 6t) dt$$

$$s = (t^3 - 3t^2)m \quad (1)$$

In order to determine the distance traveled in 3.5s, it is necessary to investigate the path of motion.

If we consider a graph of the velocity function, Fig. E-1.5b,

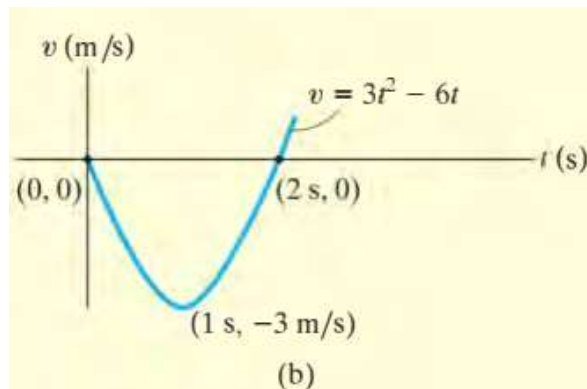


Fig. E-1.5b

then it reveals that for $0 < t < 2$ s the velocity is negative, which means the particle is traveling to the left, and for $t > 2$ s the velocity is positive, and hence the particle is traveling to the right. Also, note that $v = 0$ at $t = 2$ s. The particle's position when $t = 0$, $t = 2$ s, and $t = 3.5$ s can now be determined from Eq. 1. This yields

$$s|_{t=0} = 0 \quad s|_{t=2\text{ s}} = -4.0\text{ m} \quad s|_{t=3.5\text{ s}} = 6.125\text{ m}$$

The path is shown in Fig. 12–6a. Hence, the distance traveled in 3.5 s is

$$s_T = 4.0 + 4.0 + 6.125 = 14.125\text{ m} = 14.1\text{ m} \quad \text{Ans.}$$

Velocity. The *displacement* from $t = 0$ to $t = 3.5\text{ s}$ is

$$\Delta s = s|_{t=3.5\text{ s}} - s|_{t=0} = 6.125\text{ m} - 0 = 6.125\text{ m}$$

and so the average velocity is

$$v_{\text{avg}} = \frac{\Delta s}{\Delta t} = \frac{6.125\text{ m}}{3.5\text{ s} - 0} = 1.75\text{ m/s} \rightarrow \quad \text{Ans.}$$

The average speed is defined in terms of the *distance traveled* s_T . This positive scalar is

$$(v_{\text{sp}})_{\text{avg}} = \frac{s_T}{\Delta t} = \frac{14.125\text{ m}}{3.5\text{ s} - 0} = 4.04\text{ m/s} \quad \text{Ans.}$$

Note: In this problem, the acceleration is $a = dv/dt = (6t - 6)\text{ m/s}^2$, which is not constant.

Problem Set 1.1

1. A car starts from rest and with constant acceleration achieves a velocity of 15 m/s when it travels a distance of 200 m. Determine the acceleration of the car and the time required.
2. A particle is moving along a straight line with the acceleration $a = (12t - 3t^{1/2})\text{ m/s}^2$, where t is in seconds. Determine the velocity and the position of the particle as a function of time. When $t = 0$, $v = 0$ and $s = 15\text{ m}$.
3. A car has an initial speed of 25 m/s and a constant deceleration of 3 m/s^2 . Determine the velocity of the car when $t = 4\text{ s}$. What is the displacement of the car during the 4-s time interval? How much time is needed to stop the car?
4. A particle travels along a straight line with a velocity $v = (12 - 3t^2)\text{ m/s}$, where t is in seconds. When $t = 1\text{ s}$, the particle is located 10 m to the left of the origin. Determine the acceleration when $t = 4\text{ s}$, the displacement from $t = 0$ to $t = 10\text{ s}$, and the distance the particle travels during this time period.

5. A particle travels along a straight-line path such that in 4 s it moves from an initial position $s_A = -8m$ to a position $s_B = +3m$. Then in another 5 s it moves from s_B to $s_C = -6m$. Determine the particle's average velocity and average speed during the 9-s time interval.